HAOYU HAN

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EDUCATION

Harvard University *09 / 2024 - present*
Ph.D. in Applied Mathematics. Working on reinforcement learning, optimization and semi-definite programming.

University of Science and Technology of China (USTC) *09 / 2020 - 07 / 2024*
B.S. in Information and Computational Science (Computational Math)

PUBLICATIONS

[1] **Haoyu Han**, Heng Yang. **Building Rome with Convex Optimization**. ***Robotics: Science and Systems (RSS) 2025 Best System Paper Award*** (Link)

[2] **Haoyu Han**, Heng Yang. **Non-Uniform Noise-to-Signal Ratio in the REINFORCE Policy-Gradient Estimator**. *arXiv; under review at International Conference on Machine Learning (ICML)* (Link)

[3] Yulin Li, **Haoyu Han**, Shucheng Kang, Jun Ma, Heng Yang. **On the Surprising Robustness of Sequential Convex Optimization for Contact-Implicit Motion Planning**. *Robotics: Science and Systems (RSS) 2025* (Link)

[4] Shucheng Kang, **Haoyu Han**, Antoine Groudiev, Heng Yang. **Factorization-free Orthogonal Projection onto the Positive Semidefinite Cone with Composite Polynomial Filtering**. *arXiv; under review at SIAM Journal on Optimization (SIOPT)* (Link)

[5] **Haoyu Han**, Heng Yang. **On the Nonsmooth Geometry and Neural Approximation of the Optimal Value Function of Infinite-Horizon Pendulum Swing-up**. *Learning for Dynamics and Control (L4DC) 2024* (Link).

AWARDS

Best System Paper Award, Robotics: Science and Systems (RSS) *06 / 2025*
China National Scholarship (Ministry of Education of China) *09 / 2022*
China Optics Valley Scholarship *09 / 2023*

RESEARCH EXPERIENCE

Non-uniform NSR for Policy-Gradient Estimators [2]
Keywords: Reinforcement Learning, Policy Gradient; Stochastic Gradient Descent Harvard University
Individual Contributor *Advisor: Prof. Heng Yang* *Sept 2025 - present*

· Developed a noise-to-signal ratio (NSR) perspective for the REINFORCE estimator, explaining late-stage slowdown/instability through NSR growth driven by shrinking policy covariance and near bang-bang action distributions.
· Derived exact finite-horizon NSR characterizations for (i) linear systems with Gaussian linear policies and (ii) polynomial systems with polynomial-Gaussian policies, enabling precise diagnostics of gradient estimator quality across parameters.

Convex optimization for Structure-from-Motion (SfM) [1]

Keywords: SfM; Foundation models; Bundle adjustment; CUDA Harvard University

Individual Contributor *Advisor: Prof. Heng Yang* *May 2024 - Dec 2024*

- Proposed a scaled bundle adjustment formulation that lifts 2D keypoint measurements to 3D with learned depth from foundation models, solving it to global optimality with Semi-definite programming relaxations.
- Wrote CUDA solver dubbed XM and built an SfM pipeline with XM as the optimization engine; showed XM-SfM compares favorably in reconstruction quality while being significantly faster, more scalable, and initialization-free.

Sequential Convex Optimization for Contact-Implicit Planning [3]

Keywords: Contact-implicit planning; Numerical optimization; Trajectory optimization Harvard University

Major Contributor *Advisor: Prof. Heng Yang* *Sep 2024 - Dec 2024*

- Developed a solver dubbed CRISP improving speed and robustness for contact-implicit motion planning where Karush–Kuhn–Tucker (KKT) systems may fail; proved sufficient conditions for convergence to first-order stationary points of a merit function.
- Released a high-performance C++ implementation of CRISP with a generic nonlinear programming interface.

Computation and Analysis on Optimal Control of a Pendulum [5]

Keywords: Optimal control; Pontryagin's maximum principle; Value function Harvard University

Individual Contributor *Advisor: Prof. Heng Yang* *Aug 2023 - Dec 2023*

- Computed the cost-to-go function of the pendulum with error under 10^{-4}, with and without control constraints, using a novel contour-line methodology and Pontryagin's maximum principle (PMP).
- Discovered a non-smooth spiral in the cost-to-go function, calculated its geometry, and proved existence using symmetry and ODE theory.

REVIEW EXPERIENCE

Reviewer of The International Journal of Robotics Research (IJRR), IEEE Robotics and Automation Letters (RAL), Transactions on Machine Learning Research (TMLR), Transactions on Robotics (TRO), International Conference on Machine Learning (ICML).

TEACHING

Teaching Assistant, Harvard ES/AM 158: Introduction to Optimal Control and Reinforcement Learning *2025*

SKILLS

Computer Languages	C, C++, C#, Python, CUDA
Tools	LaTeX, Matlab, Mathematica, PyTorch, Unity, Qt, OpenCV, Markdown